HAZARD AND OPERABILITY STUDY

Hexane Storage Tank T-301

Narrative-focused experimental variant

SYNTHETIC DOCUMENT – FOR EXPERIMENTAL USE ONLY

This document is derived from a semi-synthetic dataset. It is not an actual HAZOP workshop record and must not be used for design, operation, safety assessment, permitting, or real-world decisions.

Document number	SYN-HAZOP-T301-001-B
Revision	B – narrative-focused variant
Data source	T301_HAZOP_CCPS2001_style.csv (94 records)
Study basis	Same source records and provenance as Version A
Purpose	NLP extraction and contextual relationship experiments

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1 Purpose and experimental design

This version preserves the scenario content and source classifications of Version A, but presents the scenario information as short engineering narratives rather than a worksheet matrix. The wording is a controlled, template-based rendering of the existing fields; it does not add independently verified process facts.

The document contains the same 94 scenarios: 40 reconstructed scenario chains (P) and 54 synthetic continuations (S). The source distinction is retained for every scenario. All scenarios remain unranked, and safeguard installation status is not established by this document.

2 Process context

Tank T-301 stores liquid hexane. The study treats the tank and associated inlet, outlet, vent, drain and transfer equipment as one node. Its design intent is to receive and store liquid hexane while maintaining containment and allowing controlled transfer. The upstream unloading process and downstream consumer are not studied in detail. Inter-node propagation is outside scope.

The source report describes the tank as outdoors and normally about half full. It identifies a level transmitter and level indicator controller in the level loop, and a transfer pump 4-41 supplying a downstream process that is not further specified. The conservation vent and dike shown on the reference P&ID are outside the modelled node.

3 Reading conventions

Each scenario is presented in prose. “P” means reconstructed from scenario chains reported by Single et al. (2020); it does not mean a verbatim transcription of a source worksheet. “S” means a synthetic continuation. Safeguard concepts in reconstructed records are source-listed concepts whose installation status is unverified. Synthetic records do not establish existing safeguards. “Not assessed” means no likelihood or severity ranking is provided.

4 Narrative scenario records

T301-001 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Excessive inflow. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-002 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Excessive inflow. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone,

overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-003 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Closed outlet valve; Higher-level contributing causes: Operational error, valve failure. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-004 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Closed outlet valve; Higher-level contributing causes: Operational error, valve failure. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-005 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Malfunction flow controller. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-006 – High level

This is a reconstructed scenario (source class P). The reported cause information is: Malfunction flow controller. The associated process outcome is recorded as: Reported effect: Overfilling; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, overfill protection (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-007 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Blocked inflow line; Higher-level contributing causes: Deposition of impurities. The associated process outcome is recorded as: Reported effect: Pump running dry; Reported consequence: Pump breakdown leads

to Production downtime. The safeguard information recorded for this scenario is: Dry run protection, level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-008 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Closed inlet valve; Higher-level contributing causes: Human error, operational error, valve failure. The associated process outcome is recorded as: Reported effect: Pump running dry; Reported consequence: Pump breakdown leads to Production downtime. The safeguard information recorded for this scenario is: Dry run protection, level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-009 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Closed inlet valve; Higher-level contributing causes: Human error, operational error, valve failure. The associated process outcome is recorded as: Reported effect: Emptying of container; Reported consequence: Production downtime. The safeguard information recorded for this scenario is: Level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-010 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Malfunction flow controller. The associated process outcome is recorded as: Reported effect: Emptying of container; Reported consequence: Production downtime. The safeguard information recorded for this scenario is: Level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-011 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Loss of inflow; Higher-level contributing causes: Operational error. The associated process outcome is recorded as: Reported effect: Pump running dry; Reported consequence: Pump breakdown leads to Production downtime. The safeguard information recorded for this scenario is: Dry run protection, level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-012 – Low level

This is a reconstructed scenario (source class P). The reported cause information is: Loss of inflow; Higher-level contributing causes: Operational error. The associated process outcome is recorded as: Reported effect: Emptying of container; Reported consequence: Production downtime. The

safeguard information recorded for this scenario is: Level controller low alarm (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-013 – Other than composition

This is a reconstructed scenario (source class P). The reported cause information is: Contaminants. The associated process outcome is recorded as: Reported effect: Poor product quality; Reported consequence: No safety consequence. The safeguard information recorded for this scenario is: Periodical sample taking (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-014 – Other than composition

This is a reconstructed scenario (source class P). The reported cause information is: Confusion of substances; Higher-level contributing causes: Human error. The associated process outcome is recorded as: No effect or consequence identified in source. The safeguard information recorded for this scenario is: Periodical sample taking (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-015 – Other than composition

This is a reconstructed scenario (source class P). The reported cause information is: Contamination by water; Higher-level contributing causes: Condensation air humidity, introduction of water. The associated process outcome is recorded as: Reported effect: Drainline fracture; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Water detection system, collecting basin, periodical sample taking (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-016 – Other than composition

This is a reconstructed scenario (source class P). The reported cause information is: Contamination by water; Higher-level contributing causes: Condensation air humidity, introduction of water. The associated process outcome is recorded as: Reported effect: Drainline fracture; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Water detection system, collecting basin, define Ex protection zone (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-017 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: Mechanical seal failure of pump; Higher-level contributing causes: Improper installation, operational error. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Pump breakdown leads to Production downtime. The safeguard information recorded for this scenario

is: Collecting basin (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-018 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: Mechanical seal failure of pump; Higher-level contributing causes: Improper installation, operational error. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-019 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: Mechanical seal failure of pump; Higher-level contributing causes: Improper installation, operational error. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, explosion proof motor (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-020 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: External leakage; Higher-level contributing causes: Corrosion, erosion, leaky connections, material defect, mechanical damage. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Pump breakdown leads to Production downtime. The safeguard information recorded for this scenario is: Collecting basin (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-021 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: External leakage; Higher-level contributing causes: Corrosion, erosion, leaky connections, material defect, mechanical damage. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-022 – Elsewhere flow / loss of containment

This is a reconstructed scenario (source class P). The reported cause information is: External leakage; Higher-level contributing causes: Corrosion, erosion, leaky connections, material defect, me-

chanical damage. The associated process outcome is recorded as: Reported effect: Pool formation; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, explosion proof motor (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-023 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Blocked outflow line; Higher-level contributing causes: Deposition of impurities. The associated process outcome is recorded as: Reported effect: Crack; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-024 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Blocked outflow line; Higher-level contributing causes: Deposition of impurities. The associated process outcome is recorded as: Reported effect: Crack; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, pressure relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-025 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Blocked outflow line; Higher-level contributing causes: Deposition of impurities. The associated process outcome is recorded as: Reported effect: Rupture; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure relief valve, pressure vacuum relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-026 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Blocked outflow line; Higher-level contributing causes: Deposition of impurities. The associated process outcome is recorded as: Reported effect: Rupture; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, pressure relief valve, pressure vacuum relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-027 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Thermal expansion; Higher-level contributing causes: Abnormal hot intake, external fire, solar radiation. The associated process outcome is recorded as: Reported effect: Rupture; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure relief valve, pressure vacuum relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-028 – High pressure

This is a reconstructed scenario (source class P). The reported cause information is: Thermal expansion; Higher-level contributing causes: Abnormal hot intake, external fire, solar radiation. The associated process outcome is recorded as: Reported effect: Rupture; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, pressure relief valve, pressure vacuum relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-029 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Ambient temperature change. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure vacuum relief valve, vacuum proof design (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-030 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Ambient temperature change. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, pressure vacuum relief valve, vacuum proof design, define Ex protection zone (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-031 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Obstructed vent path; Higher-level contributing causes: Faulty installation, human error. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure vacuum relief valve, vacuum proof design (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-032 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Obstructed vent path; Higher-level contributing causes: Faulty installation, human error. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone, pressure vacuum relief valve, vacuum proof design (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-033 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Excessive fluid withdrawal; Higher-level contributing causes: Malfunction transfer pump. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin, pressure vacuum relief valve, vacuum proof design (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-034 – Low pressure

This is a reconstructed scenario (source class P). The reported cause information is: Excessive fluid withdrawal; Higher-level contributing causes: Malfunction transfer pump. The associated process outcome is recorded as: Reported effect: Collapse of enclosure; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, vacuum proof design, define Ex protection zone, pressure vacuum relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-035 – High temperature

This is a reconstructed scenario (source class P). The reported cause information is: Heat input by recirculation pump; Higher-level contributing causes: Mechanical damage. The associated process outcome is recorded as: Reported effect: Abnormal evaporation; Reported consequence: Exceeding allowable pressure. The safeguard information recorded for this scenario is: Pressure relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-036 – High temperature

This is a reconstructed scenario (source class P). The reported cause information is: Heat input by recirculation pump; Higher-level contributing causes: Mechanical damage. The associated process outcome is recorded as: Reported effect: Exceeding flash point; Reported consequence: none identified. The safeguard information recorded for this scenario is: Ignition source assessment (source-listed concept; installation unverified). The recommendation field states: None mapped from source;

domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-037 – High temperature

This is a reconstructed scenario (source class P). The reported cause information is: Abnormal heat input; Higher-level contributing causes: Abnormal hot intake, external fire, solar radiation. The associated process outcome is recorded as: Reported effect: Abnormal evaporation; Reported consequence: Exceeding allowable pressure. The safeguard information recorded for this scenario is: Pressure relief valve (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-038 – High temperature

This is a reconstructed scenario (source class P). The reported cause information is: Abnormal heat input; Higher-level contributing causes: Abnormal hot intake, external fire, solar radiation. The associated process outcome is recorded as: Reported effect: Exceeding flash point; Reported consequence: none identified. The safeguard information recorded for this scenario is: Ignition source assessment (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-039 – Low temperature

This is a reconstructed scenario (source class P). The reported cause information is: Ambient temperature change. The associated process outcome is recorded as: Reported effect: Brittle fracture; Reported consequence: Loss of primary containment leads to Fire. The safeguard information recorded for this scenario is: Collecting basin (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-040 – Low temperature

This is a reconstructed scenario (source class P). The reported cause information is: Ambient temperature change. The associated process outcome is recorded as: Reported effect: Brittle fracture; Reported consequence: Loss of primary containment leads to Formation of Ex atmosphere leads to Explosion. The safeguard information recorded for this scenario is: Collecting basin, define Ex protection zone (source-listed concept; installation unverified). The recommendation field states: None mapped from source; domain-expert review required. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-041 – High level

This is a synthetic continuation (source class S). The reported cause information is: Unloading continues after target inventory; incorrect inventory estimate; level controller fails to reduce inflow.. The associated process outcome is recorded as: Liquid hexane spills from tank opening or overflow path; flammable vapor may form; fire or flash fire possible if ignition occurs.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field

states: Review high-level alarm and independent overfill shutdown; define unloading stop procedure; verify containment capacity.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-042 – High level

This is a synthetic continuation (source class S). The reported cause information is: Inlet valve fails open; transfer pump continues running; shutdown signal fails.. The associated process outcome is recorded as: Overfill and loss of containment; potential pool fire or vapor-cloud ignition.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review fail-safe inlet isolation and independent high-high level trip.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-043 – High level

This is a synthetic continuation (source class S). The reported cause information is: Level instrument under-reads; calibration drift; impulse connection obstructed.. The associated process outcome is recorded as: Operator delays stopping inflow; overfill and release of hexane.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Provide independent level verification and proof-test critical level instruments.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-044 – High level

This is a synthetic continuation (source class S). The reported cause information is: Outlet line blocked or outlet valve closed while inflow continues.. The associated process outcome is recorded as: Inventory accumulates; tank may overfill; released hexane can create a flammable pool/vapor.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Interlock or procedural control for simultaneous inflow with unavailable outlet; add high-level alarm.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-045 – High level

This is a synthetic continuation (source class S). The reported cause information is: Transfer routed to T-301 when another tank was intended.. The associated process outcome is recorded as: Unexpected inventory accumulation and potential overfill.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use positive tank/line identification and pre-transfer lineup verification.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-046 – Low level

This is a synthetic continuation (source class S). The reported cause information is: Transfer pump continues drawing liquid; inflow is lower than outflow.. The associated process outcome is recorded as: Low inventory; downstream feed interruption; pump may run dry if suction is lost.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review low-level alarm and pump trip requirements; establish minimum operating inventory.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-047 – Low level

This is a synthetic continuation (source class S). The reported cause information is: Upstream supply interruption; inlet valve fails closed; inlet strainer blocks.. The associated process outcome is recorded as: Tank empties during continued withdrawal; transfer interruption and possible pump damage.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Add low-level alarm and verify pump protection for loss of suction.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-048 – Low level

This is a synthetic continuation (source class S). The reported cause information is: Level instrument over-reads and controller reduces or stops inflow.. The associated process outcome is recorded as: Tank inventory falls below target; downstream supply may be interrupted.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Provide independent level cross-check and instrument diagnostics.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-049 – Low level

This is a synthetic continuation (source class S). The reported cause information is: Tank, drain, flange, or connected line leaks while withdrawal continues.. The associated process outcome is recorded as: Loss of hexane containment, falling inventory, flammable vapor and potential ignition.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect tank and joints; verify leak detection, isolation, and secondary containment.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-050 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Water enters through an open connection, condensation, or incomplete draining after cleaning.. The associated process outcome is recorded as: Water contamination may affect product quality and may contribute to freezing or drain-line problems under sufficiently cold conditions.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify material identity and water content; inspect drains and prevent water ingress.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-051 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Wrong substance transferred; contaminated delivery; residue remains in tank.. The associated process outcome is recorded as: Off-spec inventory; unexpected material properties; possible incompatibility depending on contaminant.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Require delivery identification and representative incoming-material sampling.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-052 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Transfer connection or tank selection error.. The associated process outcome is recorded as: Contamination and possible incompatibility; downstream product or operation affected.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use positive identification of tank, transfer hose, and material before transfer.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-053 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Tank or line returned to service without adequate draining/clearance.. The associated process outcome is recorded as: Off-spec hexane; potential unexpected interaction with residues.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Introduce documented line/tank clearance and return-to-service checks.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-054 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Water remains after washing or enters through an open drain connection.. The associated process outcome is recorded as: Potential freezing at low temperature; drain-line blockage or fracture; possible loss of containment.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify drainage and inspect drain line condition before cold-weather operation.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-055 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Supplier quality variation or inadequate receipt checks.. The associated process outcome is recorded as: Off-spec material; process performance may change; consequences depend on impurity.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Define acceptance specification and hold material pending identity/quality verification.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-056 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Sample mislabeled, contaminated, or taken from an unrepresentative point.. The associated process outcome is recorded as: Wrong operational decision; contaminated batch may be accepted.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use sample labeling, representative sampling, and repeat testing for anomalous results.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-057 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Tank opened or vent/drain connection left improperly configured.. The associated process outcome is recorded as: Potential change in vapor-space conditions; moisture contamination; flammable atmosphere may still exist depending on conditions.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review opening/closing procedure and verify connection status after maintenance.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-058 – Other than composition

This is a synthetic continuation (source class S). The reported cause information is: Water ingress, contamination, or unsuitable storage conditions.. The associated process outcome is recorded as: Product quality deterioration; downstream operating issues.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Establish periodic inspection and sampling frequency based on material specification.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-059 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Seal wear, incorrect installation, shaft misalignment, or vibration.. The associated process outcome is recorded as: Hexane release and pool formation; flammable vapor; potential fire or explosion if ignited.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect seal condition and alignment; provide suitable leak detection and isolation.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-060 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Gasket deterioration, bolt loosening, corrosion, or poor assembly.. The associated process outcome is recorded as: Hexane release near connection; pool fire or flash fire possible after ignition.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect joints and verify gasket/material compatibility and assembly quality.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-061 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Drain valve left open, damaged seat, or accidental operation.. The associated process outcome is recorded as: Hexane discharge to surrounding area; vapor generation and potential ignition.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use valve lineup checks and suitable protection against unintended opening.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-062 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Corrosion, mechanical damage, or material defect.. The associated process outcome is recorded as: Loss of primary containment; pool formation; possible fire or environmental release.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Implement condition inspection and repair criteria appropriate to tank design.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-063 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Loose fitting, damaged tubing, or maintenance error.. The associated process outcome is recorded as: Localized hexane release and potential flammable vapor accumulation.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect instrument connections and consider isolation arrangements for small-bore fittings.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-064 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Incorrect vent arrangement or liquid carryover during overfill.. The associated process outcome is recorded as: Hexane release to atmosphere; flammable vapor cloud may form.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review vent routing and overfill scenarios with qualified process-safety personnel.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-065 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Incorrect seal or gasket installed; joint not properly tightened.. The associated process outcome is recorded as: Hexane loss of containment during return to service.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use maintenance quality checks and leak testing before return to service.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-066 – Elsewhere flow / loss of containment

This is a synthetic continuation (source class S). The reported cause information is: Vehicle or handling equipment impacts tank or connected piping.. The associated process outcome is recorded as: Rapid release of hexane; pool/vapor formation; fire/explosion potential.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Assess impact protection and emergency isolation needs.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-067 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Outlet or vent path obstructed; vapor generation continues.. The associated process outcome is recorded as: Pres-

sure rise; tank damage or rupture if design capacity is exceeded; hexane release and fire/explosion potential.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify vent/relief design and inspect for blockage; assess credible overpressure cases.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-068 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Liquid warms while the tank or connected section is isolated and has insufficient expansion space.. The associated process outcome is recorded as: Pressure rise in trapped liquid or vapor space; leakage or mechanical failure possible.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review blocked-in liquid thermal expansion and ensure appropriate protection.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-069 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: External fire heats tank contents and increases vapor generation.. The associated process outcome is recorded as: Pressure rise, tank damage, loss of containment, escalation to fire involving released hexane.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Evaluate fire exposure and tank protection with qualified engineers.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-070 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Incorrect tie-in or higher-pressure source connected to tank.. The associated process outcome is recorded as: Tank overpressure if not designed for source pressure; leakage or rupture.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify connection compatibility and prevent unintended pressurization.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-071 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Instrument reads low or control action fails to respond.. The associated process outcome is recorded as: High pressure may remain undetected until release or equipment damage.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review pressure monitoring and independent protective functions where justified.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-072 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Vent path partly obstructed by debris, icing, or maintenance configuration.. The associated process outcome is recorded as: Pressure increase during filling or warming; potential loss of containment.. The safeguard information recorded for this scenario is: None established (not assumed present). The

recommendation field states: Inspect vent paths and verify capacity for relevant operating scenarios.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-073 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Tank is filled beyond intended operating inventory.. The associated process outcome is recorded as: Liquid expansion or continued inflow can cause release; tank pressure effects depend on configuration.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review maximum fill basis and independent overfill prevention.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-074 – High pressure

This is a synthetic continuation (source class S). The reported cause information is: Unapproved connection or component changes tank pressure behavior.. The associated process outcome is recorded as: Overpressure or leakage under an unintended pressure source.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Apply management of change and verify pressure ratings after modifications.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-075 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Tank cools while vapor condenses and make-up gas/venting is insufficient.. The associated process outcome is recorded as: Vacuum develops; tank may buckle or collapse if not vacuum-rated.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify vacuum protection and cooldown procedure against tank design.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-076 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Vent cannot admit sufficient make-up gas as liquid is withdrawn.. The associated process outcome is recorded as: Vacuum develops; tank deformation or collapse may cause loss of containment.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect vent path and verify vacuum relief capacity for withdrawal.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-077 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Rapid cooling condenses vapor in a tank with limited make-up gas.. The associated process outcome is recorded as: Vacuum loading and potential tank collapse.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Assess condensation/cooldown case and vacuum protection.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-078 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Transfer pump removes liquid faster than make-up gas enters.. The associated process outcome is recorded as: Vacuum condition; possible structural damage if tank is not designed for it.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review maximum withdrawal rate and venting capacity.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-079 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Tank is isolated during cooldown after hot cleaning or steam-out.. The associated process outcome is recorded as: Vacuum formation and possible enclosure collapse.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Use approved cleaning/cooldown procedure and verify vacuum protection.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-080 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Blanketing/make-up gas source is unavailable where such a system is used.. The associated process outcome is recorded as: Vacuum may develop during withdrawal or cooling; air ingress may alter vapor-space conditions.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: If blanketing is part of the design, alarm on supply failure and verify independent vacuum protection.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-081 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Pressure instrument is biased or blocked.. The associated process outcome is recorded as: Operator may fail to recognize vacuum condition.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect and test pressure instruments; use independent verification for critical indication.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-082 – Low pressure

This is a synthetic continuation (source class S). The reported cause information is: Tank and vent/make-up paths isolated while contents cool.. The associated process outcome is recorded as: Vacuum formation and possible structural damage.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review isolation line-up and provide a safe cooldown path.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-083 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Prolonged solar exposure raises stored liquid temperature.. The associated process outcome is recorded as: Higher vapor generation and pressure; potential exceedance of tank/vent capacity depending on conditions.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Assess ambient heat exposure and verify pressure/venting design.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-084 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Tank exposed to nearby fire or radiant heat.. The associated process outcome is recorded as: Rapid heating and vapor generation; pressure rise; possible loss of containment.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review fire exposure, emergency response, and applicable tank protection.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-085 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Unexpectedly hot liquid transferred into tank.. The associated process outcome is recorded as: Temperature excursion, increased vapor generation, possible pressure rise.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify incoming material temperature where relevant and define transfer acceptance conditions.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-086 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Mechanical fault or prolonged recirculation adds heat to stored liquid.. The associated process outcome is recorded as: Increased evaporation and vapor pressure; potential pressure-control challenge.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Monitor pump condition and evaluate heat input during abnormal operation.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-087 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Transfer or recirculation occurs under unintended thermal conditions.. The associated process outcome is recorded as: Unexpected vapor generation and pressure increase.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Review operating sequence and temperature monitoring needs.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-088 – High temperature

This is a synthetic continuation (source class S). The reported cause information is: Temperature instrument under-reads actual liquid temperature.. The associated process outcome is recorded as:

Abnormal heating may go unnoticed; delayed response to rising vapor generation.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify temperature indication and calibration; use independent checks where warranted.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-089 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Cold weather cools the tank or connected drain line.. The associated process outcome is recorded as: Material/metal may be exposed to low-temperature conditions; water in drains may freeze; brittle fracture is possible if material suitability is inadequate.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Check minimum design metal temperature/material suitability and protect vulnerable drains.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-090 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Colder-than-intended material enters tank.. The associated process outcome is recorded as: Thermal excursion; potential material embrittlement concerns depending on design and material.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Verify temperature compatibility and review material design limits.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-091 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Cold utility contacts a process boundary due to exchanger or line failure.. The associated process outcome is recorded as: Localized cooling; possible brittle fracture if below material limits.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect heat-transfer boundaries and assess low-temperature material suitability.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-092 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Temperature instrument reads higher than actual temperature.. The associated process outcome is recorded as: Cold condition remains undetected; freezing or embrittlement concerns may be missed.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Calibrate temperature instrumentation and assess independent monitoring.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-093 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Water remains in drain or low point and ambient temperature falls.. The associated process outcome is recorded as: Ice may obstruct or damage drain line; leakage may follow if the line fractures.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field

states: Verify drainage and inspect/maintain low-point drains for the expected environment.. The source report does not assign a risk ranking; the scenario remains not assessed.

T301-094 – Low temperature

This is a synthetic continuation (source class S). The reported cause information is: Insulation or weather protection is damaged or absent.. The associated process outcome is recorded as: Cold spots; possible embrittlement or water freezing depending on materials and contents.. The safeguard information recorded for this scenario is: None established (not assumed present). The recommendation field states: Inspect weather protection and verify material suitability for ambient conditions.. The source report does not assign a risk ranking; the scenario remains not assessed.

5 Limitations

This is a synthetic experimental document generated from structured records. It does not establish the physical validity of every cause–effect chain. The 54 synthetic continuations are generated examples, not verified plant events. The 40 reconstructed scenarios are reconstructions of published scenario chains, not original workshop records. Expert review and authentic industrial reports are required before drawing conclusions about real-world HAZOP performance.

6 References

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[3] IEC 61882:2016. *Hazard and operability studies (HAZOP studies) – Application guide*.